Therefore, the equation of the required line is $y - y_1 = m(x - x_1)$ (Point slope form)

Thinking Corner

Is it possible to express, the equation of a straight line in slope-intercept form, when it is parallel to Y axis?

Example 5.21 Find the equation of a line passing through the point $(3, -4)$ and having slope $\frac{-5}{7}$

Solution Given, $(x_1, y_1) = (3, -4)$ and $m = \frac{-5}{7}$

The equation of the point-slope form of the straight line is $y - y_1 = m(x - x_1)$

we write it as $y + 4 = -\frac{5}{7}(x - 3)$

$\Rightarrow \quad 5x + 7y + 13 = 0$

Example 5.22 Find the equation of a line passing through the point $A(1,4)$ and perpendicular to the line joining points $(2,5)$ and $(4,7)$.

Solution

Let the given points be $A(1,4)$, $B(2,5)$ and $C(4,7)$.

Slope of line $BC = \frac{7-5}{4-2} = \frac{2}{2} = 1$

Let m be the slope of the required line.

Since the required line is perpendicular to BC,

$$m \times 1 = -1$$
$$m = -1$$

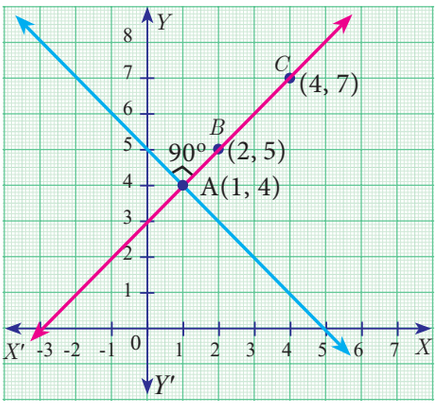


Fig. 5.33

The required line also pass through the point $A(1,4)$.

The equation of the required straight line is $y - y_1 = m(x - x_1)$

$$y - 4 = -1(x - 1)$$
$$y - 4 = -x + 1$$

we get, $x + y - 5 = 0$

5.5.6 Two Point form

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be two given distinct points. Slope of the straight line passing through these points is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$, $(x_2 \neq x_1)$.

From the equation of the straight line in point slope form, we get

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

Hence, $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$ (is the equation of the line in two-point form)

Example 5.23 Find the equation of a straight line passing through $(5, -3)$ and $(7, -4)$.

Solution The equation of a straight line passing through the two points (x_1, y_1) and (x_2, y_2) is

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